Payments, Patterns & Performance: A Data-Driven Case Study on Olist Brazilian E-Commerce

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Introduction & Objectives

The Brazilian e-commerce sector has experienced rapid expansion, with platforms like Olist enabling sellers to reach a broad customer base. This case study examines the Olist Brazilian E-Commerce Public Dataset, which includes transactions from 2016 to 2018, to uncover insights that inform business and payment strategies.

The study focuses on three key areas:

1. **Revenue and Demand**

   To identify growth trends across time periods and product categories and highlight underperforming segments.

2. **Customer Experience**

   To assess how operational factors such as delivery delays and cancellations affect customer satisfaction.

3. **Payment Behavior**

   To analyze how payment methods influence transaction value and overall customer experience.

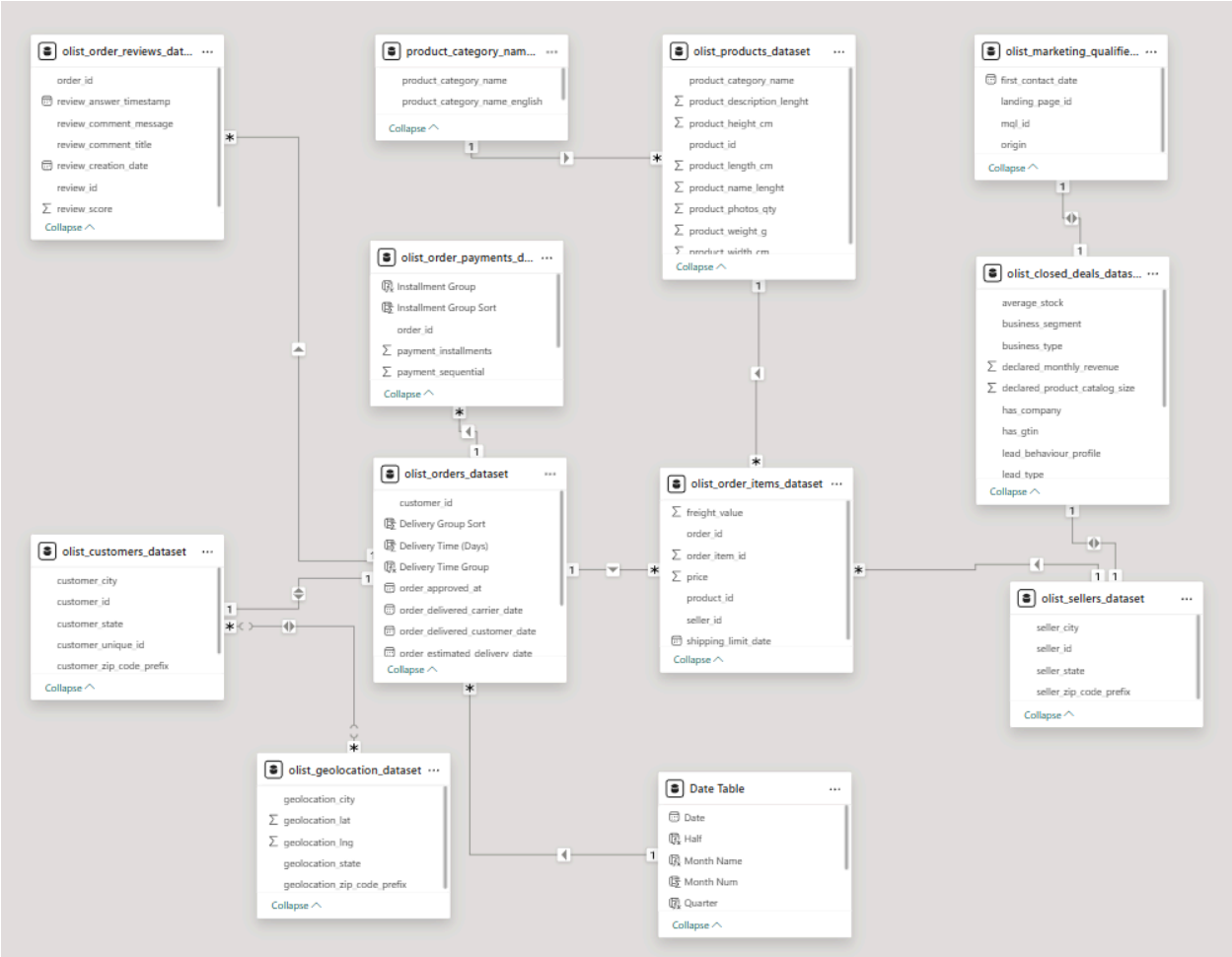
These objectives are particularly relevant for organizations focused on payments, where understanding customer transaction behavior is essential for optimizing financial and product strategies.

Methodology

Data Source

The dataset was obtained from Kaggle and consists of 11 interconnected CSV files containing information on orders, customers, sellers, products, payments, and reviews. An Entity Relationship Diagram (ERD) and Data Dictionary were also provided as part of the case study instructions by PayMongo, which were used to guide schema design, table relationships, and data interpretation.

Data model (Power BI)



Data Preparation and Cleaning

Data Insertion in Microsoft SQL Server

SQLQuery1.s...\Riley (69))*

```sql
-- ============================================================
-- Olist Brazilian E-Commerce Dataset - MS SQL Server Import
-- Schema based on: dbdiagram.io ERD (full dataset)
-- CSV Path: C:\Users\Riley\Documents\Denzel\Case Study\2024\Dataset
-- ============================================================

USE master;
GO

IF NOT EXISTS (SELECT name FROM sys.databases WHERE name = 'OlistEcommerce')
BEGIN
    CREATE DATABASE OlistEcommerce;
END
GO

USE OlistEcommerce;
GO

-- ============================================================
-- DROP TABLES in FK-safe order
-- ============================================================
IF OBJECT_ID('dbo.olist_closed_deals_dataset',               'U') IS NOT NULL DROP TABLE dbo.olist_closed_deals_dataset;
IF OBJECT_ID('dbo.olist_marketing_qualified_leads_dataset',   'U') IS NOT NULL DROP TABLE dbo.olist_marketing_qualified_leads_dataset;
IF OBJECT_ID('dbo.olist_order_payments_dataset',              'U') IS NOT NULL DROP TABLE dbo.olist_order_payments_dataset;
IF OBJECT_ID('dbo.olist_order_reviews_dataset',               'U') IS NOT NULL DROP TABLE dbo.olist_order_reviews_dataset;
IF OBJECT_ID('dbo.olist_order_items_dataset',                 'U') IS NOT NULL DROP TABLE dbo.olist_order_items_dataset;
IF OBJECT_ID('dbo.olist_orders_dataset',                      'U') IS NOT NULL DROP TABLE dbo.olist_orders_dataset;
IF OBJECT_ID('dbo.olist_products_dataset',                    'U') IS NOT NULL DROP TABLE dbo.olist_products_dataset;
IF OBJECT_ID('dbo.product_category_name_translation',         'U') IS NOT NULL DROP TABLE dbo.product_category_name_translation;
IF OBJECT_ID('dbo.olist_sellers_dataset',                     'U') IS NOT NULL DROP TABLE dbo.olist_sellers_dataset;
IF OBJECT_ID('dbo.olist_customers_dataset',                   'U') IS NOT NULL DROP TABLE dbo.olist_customers_dataset;
IF OBJECT_ID('dbo.olist_geolocation_dataset',                 'U') IS NOT NULL DROP TABLE dbo.olist_geolocation_dataset;
GO

-- ============================================================
-- 1. GEOLOCATION  (no FK dependencies)
--    Note: geolocation_zip_code_prefix has duplicate values
--    in the raw CSV so no single-column PK is enforced here.
-- ============================================================
CREATE TABLE dbo.olist_geolocation_dataset (
    geolocation_zip_code_prefix INT          NOT NULL,
    geolocation_lat             FLOAT        NOT NULL,
    geolocation_lng             FLOAT        NOT NULL,
    geolocation_city            VARCHAR(100),
    geolocation_state           CHAR(2)
);
GO

-- ============================================================
```

100 % No issues found

Connected. (1/1) DESKTOP-2KS0B12\SQLEXPRESS ... DESKTOP-2KS

Data was imported into Microsoft SQL Server using BULK INSERT operations. Several preprocessing steps were performed:

- Removal of embedded newlines in text fields using Python
- Conversion of boolean text values into BIT format
- Handling null and blank values using TRY_CAST and NULLIF
- Managing duplicate geolocation entries

- Grouping missing product categories under “Others”

These steps ensured data consistency and prevented errors during analysis.

Python Script Data Cleaning

```python
import csv
import os

DATASET_PATH = r"C:\Users\Riley\Documents\Denzel\Case Study\2024\Dataset"

# ------------------------------------------------------
# FIX 1: Clean olist_order_reviews_dataset.csv
# Replaces embedded newlines inside fields with a space
# ------------------------------------------------------

input_reviews  = os.path.join(DATASET_PATH, "olist_order_reviews_dataset.csv")
output_reviews = os.path.join(DATASET_PATH, "olist_order_reviews_dataset_clean.csv")

print("Cleaning reviews CSV...")

rows_read    = 0
rows_written = 0

with open(input_reviews, "r", encoding="utf-8") as infile, \
     open(output_reviews, "w", encoding="utf-8", newline="") as outfile:

    reader = csv.reader(infile)
    writer = csv.writer(outfile, quoting=csv.QUOTE_MINIMAL)

    for i, row in enumerate(reader):
        rows_read += 1

        # Replace embedded newlines/carriage returns inside each field
        cleaned = [
            field.replace("\r\n", " ").replace("\r", " ").replace("\n", " ")
            for field in row
        ]
        writer.writerow(cleaned)
        rows_written += 1

print(f"  Read:    {rows_read:,} rows (including header)")
print(f"  Written: {rows_written:,} rows")
print(f"  Output:  {output_reviews}")
print()
print("Done! Now run olist_final_load.sql in SQL Server.")
```

Data Analysis

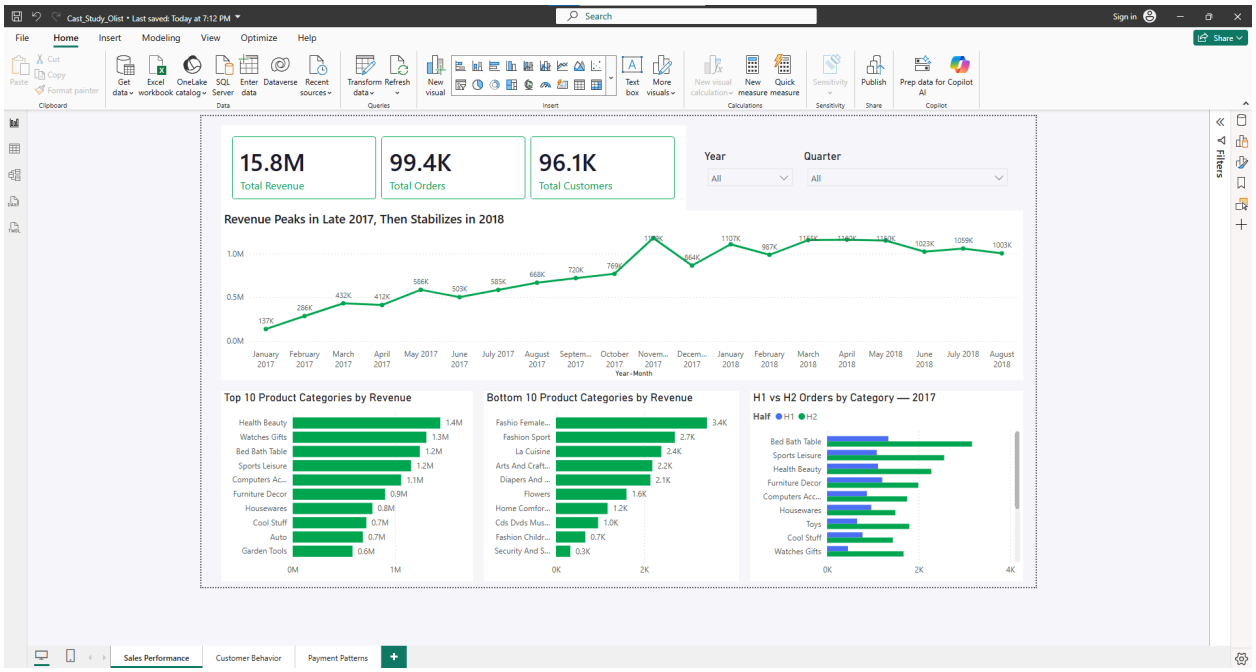
SQL Server Management Studio (SSMS) was used for querying and analysis. Techniques included:

- Aggregate functions
- Conditional logic
- Common Table Expressions (CTEs)
- Subqueries and window functions

The analysis focused on sales trends, customer satisfaction metrics, and payment distributions.

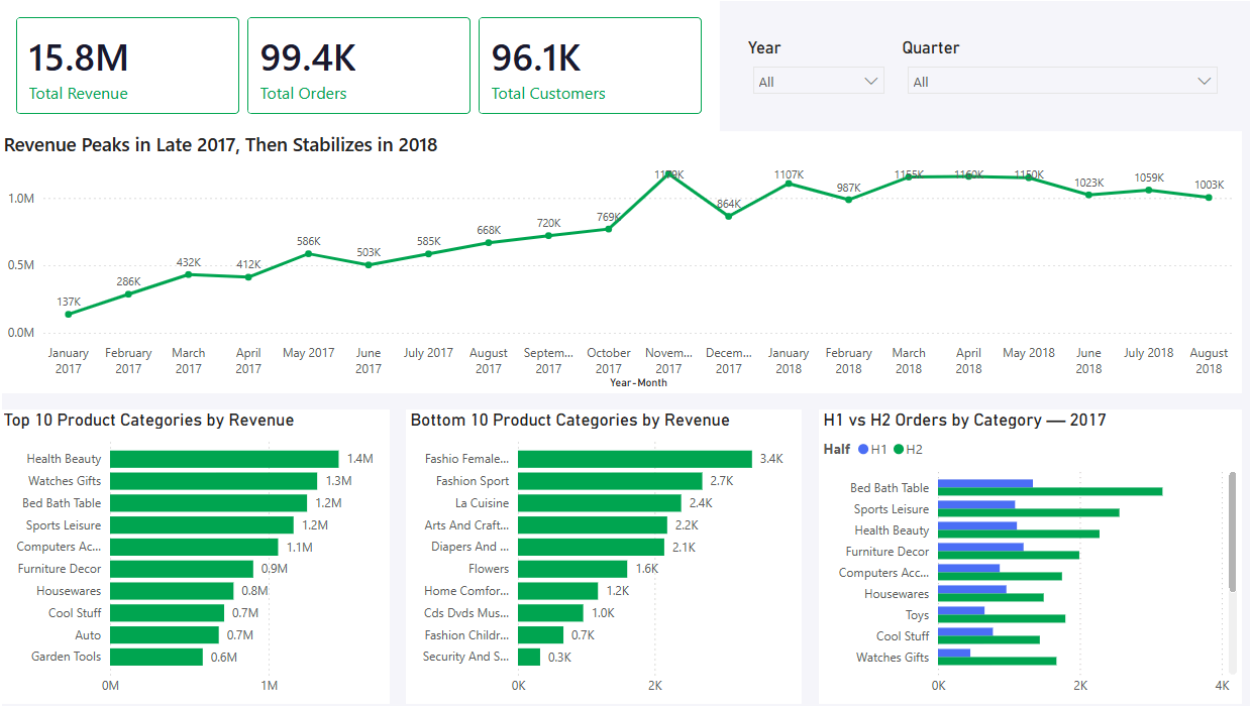
Data Visualization

Microsoft Power BI was used to create dashboards and visualizations. Excel was used for validation and supplementary analysis.

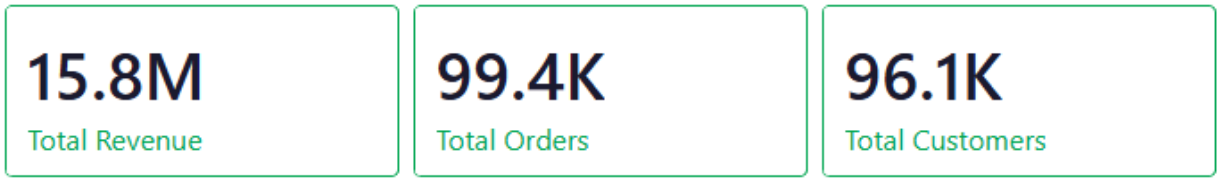


Results

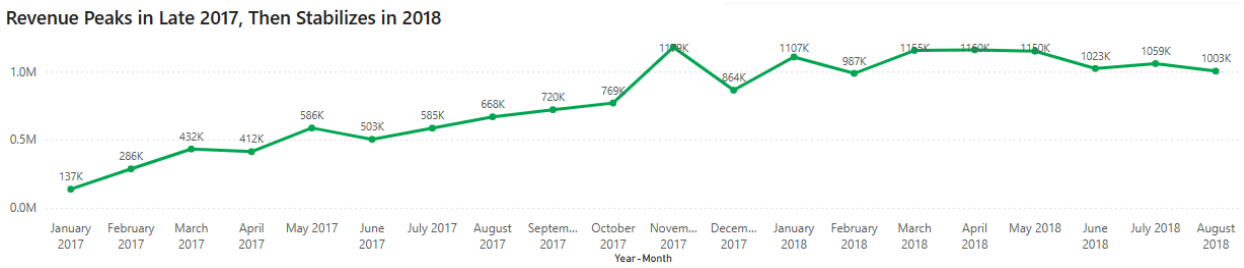
Sales Performance



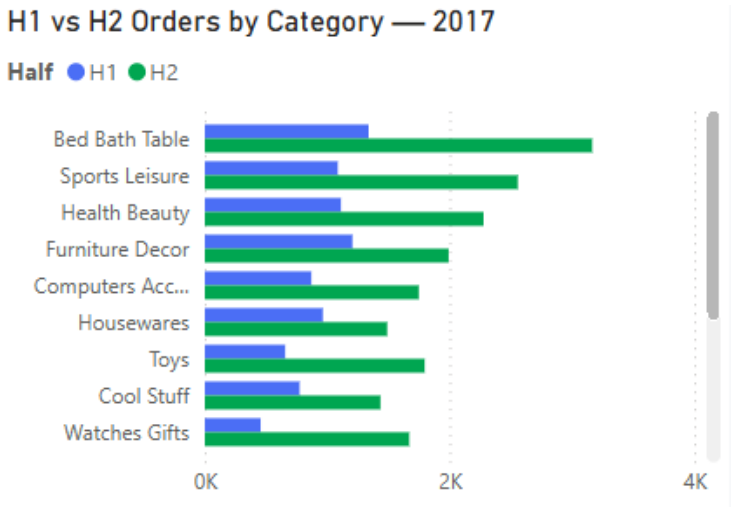
Olist experienced remarkable growth over the analysis period. Total revenue across all orders reached R$15.8M, driven by 99,400 orders from 96,100 unique customers across 27 Brazilian states.



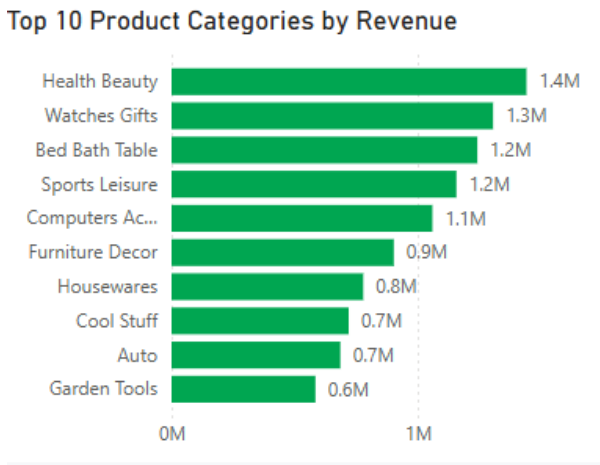
Revenue grew approximately 10x from September 2016 to November 2017, peaking at R$1.2M in a single month before stabilizing at a consistent R$800K–R$1M monthly range through mid-2018. This growth trajectory reflects the rapid expansion of Brazilian e-commerce during this period and Olist's growing marketplace presence.

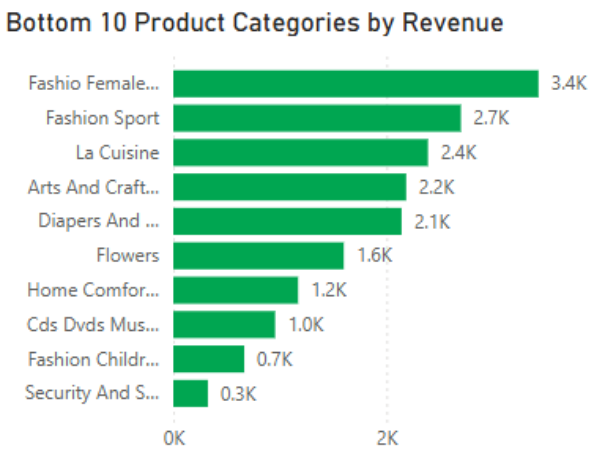


A clear seasonality pattern emerged across product categories — H2 consistently outperformed H1 in every single category analyzed in 2017. Bed, bath & table led with 1,524 orders in H1 growing to 3,692 in H2 — a 142% increase. Health & beauty, sports & leisure, and computers & accessories followed as the top revenue-generating categories overall. This seasonality is likely driven by Brazilian holiday shopping behavior in Q4, particularly around Black Friday and Christmas.

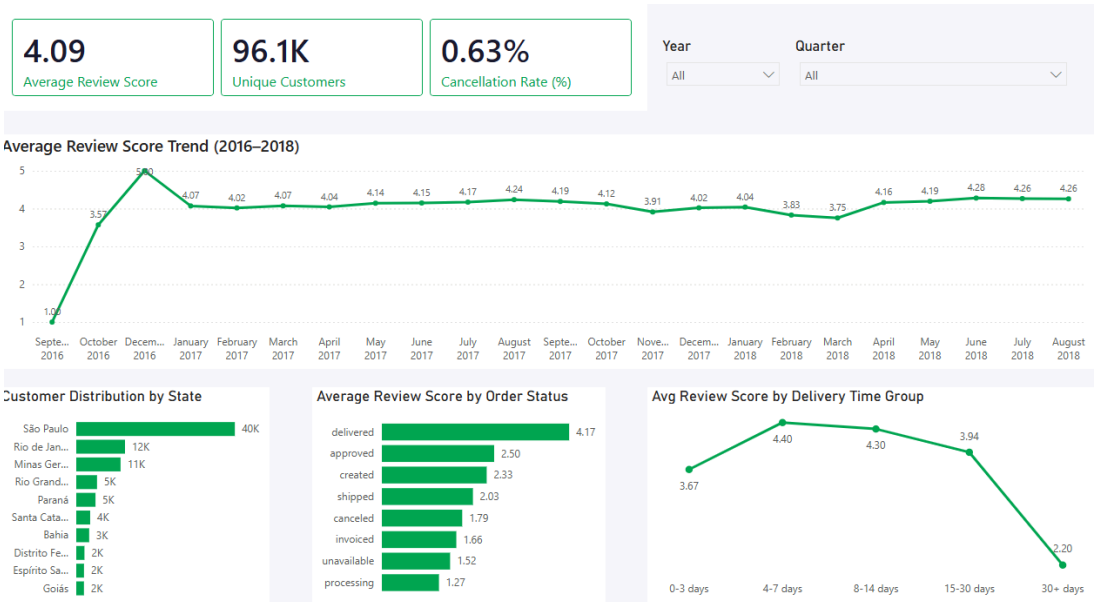


The bottom performing categories — including flowers, security & services, and arts & craftsmanship — generated minimal order volumes, suggesting either low market demand or insufficient seller representation in these segments.

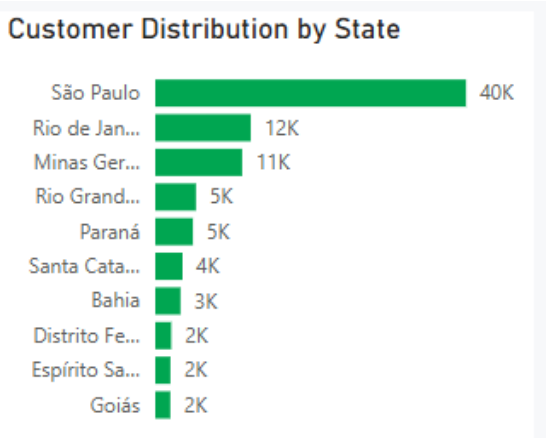




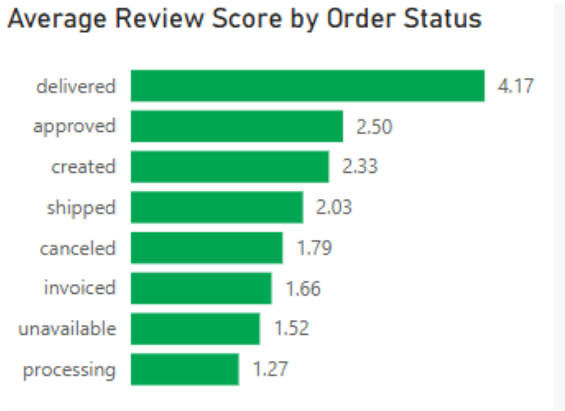
Customer Behavior



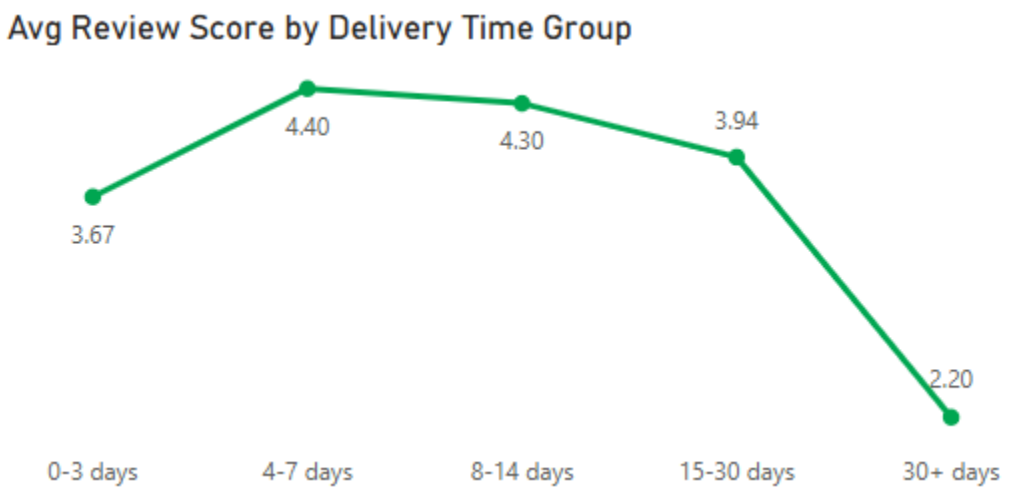
Olist's customer base is heavily concentrated in the Southeast region of Brazil. São Paulo alone accounts for over 40,000 unique customers — nearly half the entire customer base — followed by Rio de Janeiro at 12,000 and Minas Gerais at 11,000. This geographic concentration reflects broader economic patterns in Brazil where purchasing power is highest in the Southeast.



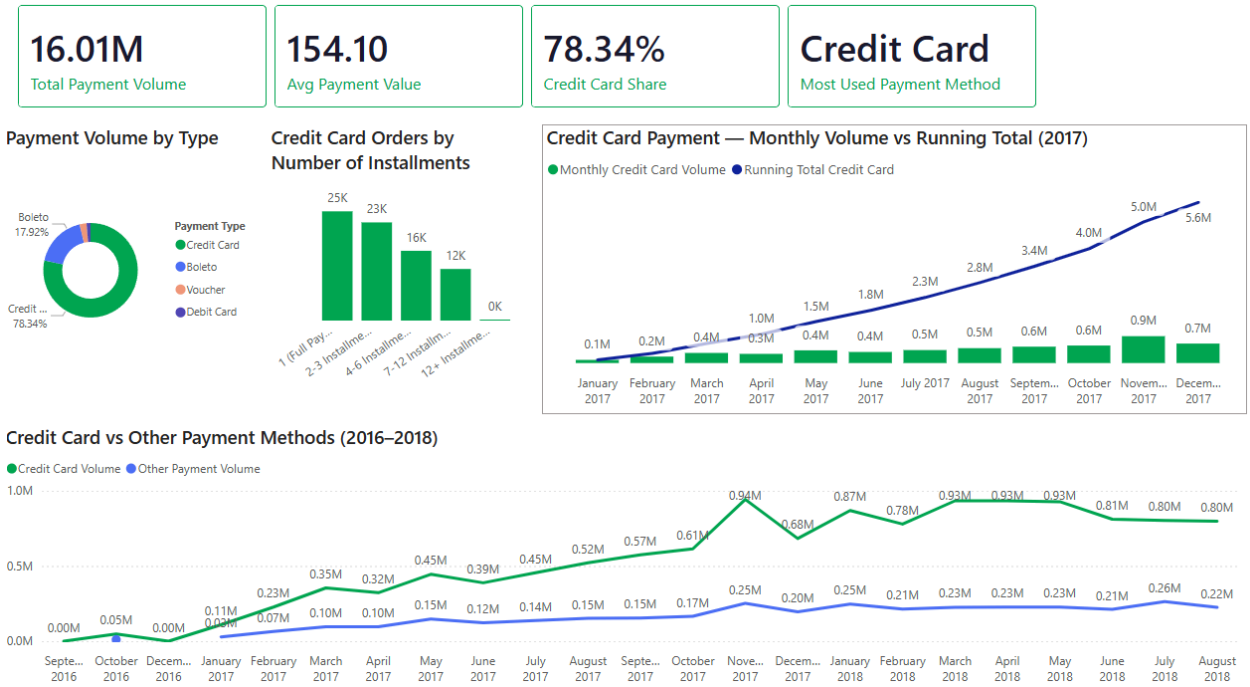
Customer satisfaction averaged 4.09 out of 5 across all orders — a generally positive score, though with significant variation depending on order outcome and delivery performance. Orders with a delivered status averaged 4.17, while canceled orders dropped to 1.79 and unavailable orders scored just 1.52 — confirming that order fulfillment is the primary driver of customer satisfaction.



The relationship between delivery time and satisfaction is unambiguous. Orders delivered within 0–7 days averaged above 4.40, while orders taking 30+ days dropped to 2.20. This near-linear relationship between delivery speed and review score is the single most actionable finding in the dataset — every additional week of delivery time meaningfully erodes customer satisfaction.

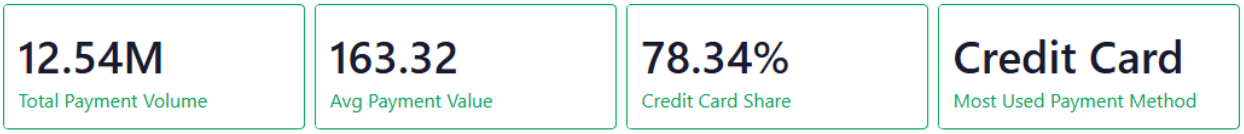


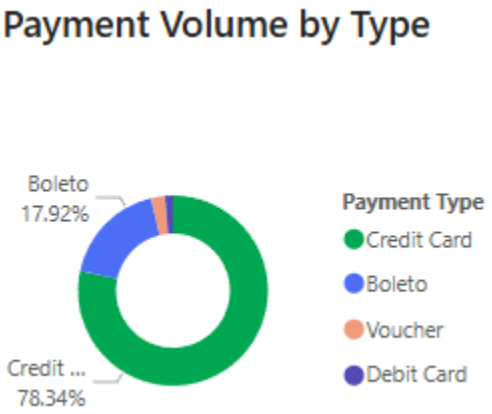
Payment Patterns



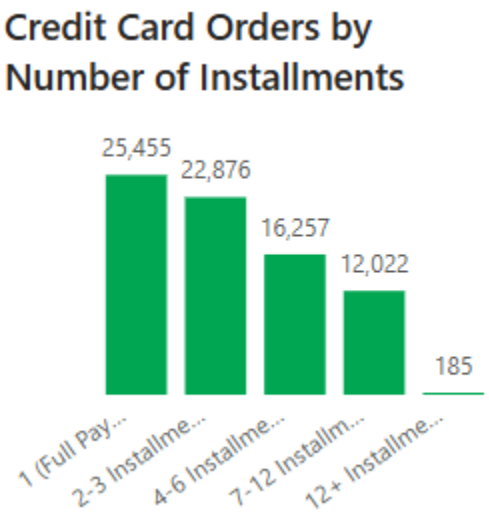
Credit card is the overwhelmingly dominant payment method on the Olist platform, accounting for 78.34% of total payment volume — R$12.5M out of R$16.01M total. Boleto (a Brazilian bank slip payment method) accounts for 17.92%, while voucher and debit card together represent less than 4%.

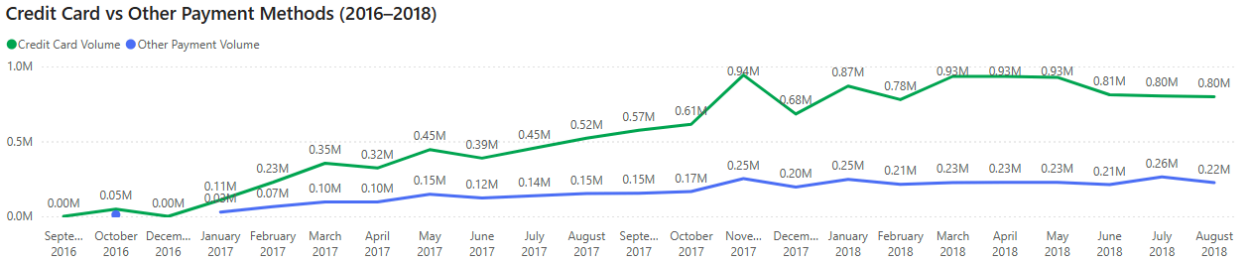
The average transaction value across all payment types was R$154.10, though this varies significantly by method. Debit card transactions averaged R$324.52 — the highest of any payment type — followed by credit card at R$231.51 and boleto at R$178.11. Voucher transactions averaged just R$64.49, reflecting their typical use for small discount redemptions.



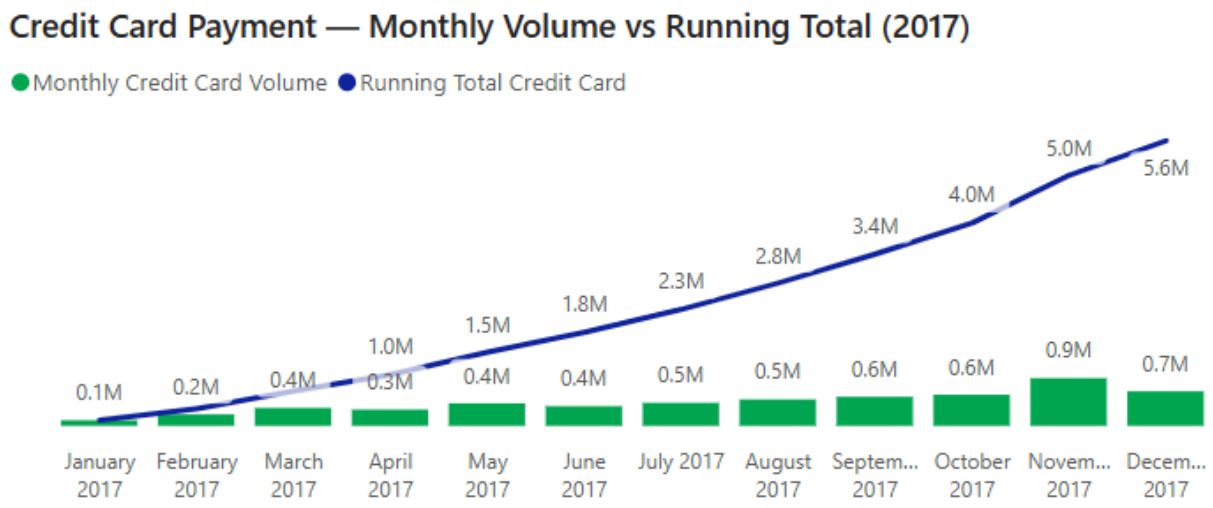


The installment analysis reveals important behavioral insights. Of all credit card transactions, 25,000 were paid in full (1 installment), 23,000 in 2–3 installments, 16,000 in 4–6 installments, and 12,000 in 7–12 installments. This distribution shows that while full payment is most common, a significant portion of Brazilian consumers — nearly 75% — prefer to split their purchases across multiple installments, reflecting the cultural prevalence of parcelamento (installment payments) in Brazilian consumer behavior.





Credit card payment volume accumulated R$5.6M in total throughout 2017, with consistent monthly contributions averaging R$470K and a notable spike in November 2017 aligned with the broader seasonal revenue peak.



Recommendations

Strategic Recommendations

1. **Capitalize on Seasonal Demand**

   Increase inventory, marketing, and logistics capacity ahead of Q4 peak periods.

2. **Improve Delivery Speed**

   Establish stricter delivery SLAs and proactively manage delays to maintain high customer satisfaction.

3. **Enhance Payment Flexibility**

   Expand installment options to align with consumer preferences and increase conversion rates.

4. **Expand Regional Market Coverage**

   Target underpenetrated regions with localized strategies to grow the customer base.

5. **Reduce Cancellations**

   Use predictive indicators to identify and mitigate high-risk orders.

Suggestions for Improvement

- Implement real-time order tracking notifications to manage customer expectations on delivery timelines
- Build a seller performance scoring system based on delivery time, cancellation rate, and review scores to surface high-quality sellers

- Introduce payment method incentives — such as cashback for debit card or boleto payments — to diversify away from credit card dependency and reduce payment processing costs.

Areas for Further Study

- **Customer lifetime value (CLV)** — the dataset shows unique customers but doesn't track repeat purchase behavior over time. A follow-up study on repeat purchase rates by state and category would strengthen retention strategy
- **Seller profitability analysis** — matching seller revenue against product weight and freight costs could identify which seller segments are most profitable
- **Marketing channel ROI** — the MQL and closed deals data shows which lead origins generate the most conversions, but a deeper funnel analysis connecting lead source to long-term seller revenue would be valuable

Conclusion

The Olist Brazilian E-Commerce dataset reveals a marketplace in strong growth mode — revenue scaled 10x from September 2016 to November 2017, with over 99,000 orders processed across 27 Brazilian states. Three findings stand out as most actionable:

First, **seasonality is real and predictable** — H2 consistently outperforms H1 across every product category, with November as the clear revenue peak. Businesses that prepare for this window will outperform those that don't.

Second, **delivery speed is the single biggest driver of customer satisfaction** — the correlation between delivery time and review score is unambiguous. Customers who receive orders within 7 days rate their experience 4.3+ out of 5, while those waiting 30+ days drop to 2.20. Logistics investment is customer satisfaction investment.

Third, **credit card dominates payments but installment behavior reveals opportunity** — the majority of credit card transactions are split across multiple installments, reflecting Brazilian consumers' preference for flexible payment structures. Payment infrastructure that supports seamless installment experiences will win in this market.